Punjab GeoGuardian Final Report

Project

Punjab GeoGuardian is an educational GeoAI prototype for monitoring possible urban expansion and farmland loss in Faisalabad District, Punjab, Pakistan.

Data and Method

The project compares Sentinel-2 satellite imagery from 2019 and 2025. Cloud-masked median composites were created using Google Earth Engine. NDVI and NDBI were calculated, followed by Random Forest land-cover classification.

The classes are:

- Other
- Vegetation / Agriculture
- Built-up / Urban

Possible conversion is defined as pixels classified as vegetation/agriculture in 2019 and built-up/urban in 2025.

Results

- Total AOI area: 5,953.79 km2
- Vegetation/agriculture area in 2019: 4,643.94 km2
- Built-up/urban area in 2025: 838.99 km2
- Estimated vegetation-to-built-up conversion: 344.84 km2
- Estimated AOI affected: 5.79%
- Validation points: 60
- Validation accuracy: 96.67%

Limitations

The results are estimates. Dry soil, seasonal crop changes, mixed pixels, small buildings, image dates, cloud masking, and training-label quality can affect classification accuracy.

Disclaimer

This project is an educational learning prototype. It is not an official government monitoring system, legal land record, or official decision-making system.